Discount Curve Estimation from Coupon Bonds: A Monte Carlo Assessment of Smoothness Regularization

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Abstract

We study the estimation of the discount and yield curves from prices of coupon-bearing bonds using a smoothness-regularized framework. Following the approach of Filipović–Pelger–Ye (Filipović et al., 2022), the discount curve is recovered by trading off weighted pricing accuracy against an economically motivated smoothness penalty, leading to a closed-form kernel ridge regression estimator (Filipović et al., 2022). This regularization addresses the intrinsic ill-posedness of recovering a continuous term structure from a sparse cross-section of bond prices with overlapping cash flows and measurement noise. We evaluate the finite-sample behavior of the estimator using Monte Carlo simulations designed to mirror the key features of Treasury bond markets: heterogeneous coupon structures, limited maturity coverage, and noisy observed prices. The simulations are conducted in a static, cross-sectional setting to isolate identification and regularization effects. Estimation and tuning are performed using cross-validation with pricing RMSE as the primary objective. As a benchmark, we implement the Nelson–Siegel–Svensson specification (Nelson and Siegel, 1987; Svensson, 1994) under the same simulation and tuning protocol. The results highlight the role of smoothness regularization in stabilizing discount-curve estimation and improving pricing accuracy. While yield-based diagnostics provide complementary information, pricing performance remains the central criterion for evaluating curve estimates derived from coupon bonds.

Keywords: discount curve estimation; yield curve; term structure of interest rates; smoothness regularization; kernel ridge regression; Monte Carlo simulation; coupon bonds; Nelson–Siegel–Svensson.

1 Introduction

The discount and yield curves are fundamental objects in empirical finance and macroeconomics. They summarize the term structure of risk-free discounting and enter directly into asset pricing, risk management, and monetary policy analysis. In practice, however, the yield curve is not directly observed. Instead, it must be inferred from the prices of coupon-bearing bonds that trade at discrete maturities and are subject to market imperfections and measurement noise.

Recovering a continuous discount curve from a finite cross-section of coupon-bond prices is an inherently ill-posed inverse problem. On any given date, the number of observed securities is small relative to the number of maturity points required to represent a smooth curve over long horizons. As a result, many discount curves can price the observed bonds nearly equally well. Without additional structure, the pricing equations alone do not identify a unique curve.

The framework developed by Filipović–Pelger–Ye (Filipović et al., 2022) formalizes the necessary trade-off between pricing fit and economic plausibility by introducing a smoothness penalty on the discount curve. This penalty reflects limits-to-arbitrage considerations: highly irregular discount curves imply extreme payoffs for dynamic trading strategies and are therefore economically implausible. Imposing smoothness regularization renders the estimation problem well-posed and leads to a closed-form solution that can be expressed as a kernel ridge regression. (Filipović et al., 2022)

This paper focuses on the finite-sample performance of this smoothness-regularized estimator in a controlled Monte Carlo environment. Rather than introducing time-series dynamics, we adopt a purely static, cross-sectional perspective that mirrors how yield curves are typically constructed in practice on a date-by-date basis. The simulation design isolates the central challenges of curve estimation from coupon bonds: overlapping cash flows, limited maturity coverage, and noisy prices.

Our evaluation proceeds in two stages. First, we assess structural properties of the estimator, examining its ability to recover the underlying discount and yield curves and to reflect differences in identification strength across maturities. Second, we benchmark pricing and yield performance against the Nelson–Siegel–Svensson specification (Nelson and Siegel, 1987; Svensson, 1994) using identical data and cross-validation procedures. Throughout, pricing accuracy is treated as the primary performance metric, reflecting both the estimation objective and the economic role of the discount curve as the pricing primitive.

2 Setup and Notation

We work in a static, cross-sectional setting and suppress time subscripts throughout. All maturities are measured in years and denoted by $x \geq 0$.

2.1 Discount curve and yields

The primitive object of interest is the discount curve $g(x)$, interpreted as the price at time zero of a zero-coupon bond paying one unit at maturity x. We impose the normalization $g(0) = 1$ and view g as a function $g : [0, \infty) \to \mathbb{R}$.

The zero-coupon yield curve $y(x)$ is defined by

$$y(x) := -\frac{1}{x}\log g(x), \qquad x > 0,$$

with the inverse relationship

$$g(x) = \exp(-x\, y(x)).$$

The discount and yield representations are interchangeable and will be used according to convenience.

2.2 Coupon bonds and pricing

We observe M coupon-bearing bonds with prices $P_1, \ldots, P_M$. Cash flows are paid on a common set of dates

$$0 < x_1 < \cdots < x_N,$$

which define the maturity grid. Let C_{ij} denote the cash flow of bond i paid at date x_j, and collect these cash flows in the matrix $C \in \mathbb{R}^{M\times N}$.

Given a candidate discount curve g, the model-implied price of bond i is

$$P_i^g = \sum_{j=1}^{N} C_{ij} g(x_j).$$

Observed prices may deviate from these fundamental values due to market imperfections and measurement noise. We write

$$P_i = P_i^g + \varepsilon_i,$$

and stack prices as $P = (P_1, \ldots, P_M)^\top$.

2.3 Smoothness and regularization

Because the number of observed bonds is limited relative to the dimensionality of the maturity grid, recovering g from prices alone is ill-posed. Following the smoothness-regularized framework, admissible discount curves are restricted to a class of sufficiently smooth functions. Smoothness is measured through a penalty that trades off variation in slope and curvature, possibly with maturity weighting.

Estimation proceeds by minimizing a penalized objective that combines weighted pricing errors with a smoothness penalty,

$$\min_g \left\{ \sum_{i=1}^{M} \omega_i (P_i - P_i^g)^2 + \lambda \|g\|^2 \right\},$$

where $\lambda > 0$ controls the strength of regularization and $\omega_i > 0$ are exogenous weights. The smoothness norm $\|\cdot\|$ induces a reproducing kernel Hilbert space structure, allowing the solution to be expressed in closed form as a kernel ridge regression estimator, as established in Theorem 1. (Filipović et al., 2022)

The weighting scheme is chosen to reflect the pricing relevance of each bond and to provide an approximate link between pricing errors and yield errors. The regularization parameter λ is selected empirically using cross-validation with pricing RMSE as the objective.

3 Core Modeling Assumptions

This section collects the modeling assumptions that underpin the discount-curve estimation problem and its kernel-ridge solution. The emphasis is on assumptions that are intrinsic to the economic and statistical structure of the problem, rather than on ad-hoc functional restrictions imposed for convenience.

3.1 Law of one price and cash-flow representation

Assumption 1 (Static valuation under no arbitrage)**.** Each coupon bond can be represented as a portfolio of zero-coupon claims at its cash-flow dates. Let C_{ij} denote the cash flow of bond i paid at date x_j. Then, for any candidate discount curve g, the fundamental value of bond i is

$$P_i^g = \sum_{j=1}^{N} C_{ij}\, g(x_j).$$

This expresses the law of one price: absent arbitrage, the portfolio that replicates the bond's cash flows must have the same value as the bond.

3.2 Market imperfections and pricing errors

Assumption 2 (Noisy observed prices)**.** Observed bond prices may deviate from their fundamental values due to market imperfections and data issues. We model this wedge as an additive pricing error,

$$P_i = P_i^g + \varepsilon_i,$$

where "market imperfections" is understood broadly to include limited liquidity, trading frictions, and measurement error. No distributional assumptions on ε_i are imposed at this stage. The goal is not to force exact pricing of all securities, but to estimate a discount curve that fits the cross-section well while remaining economically plausible.

3.3 Regularization via economically motivated smoothness

Because the set of maturities is essentially continuous while the number of observed bond prices is finite, discount-curve estimation is ill-posed without regularization.

Assumption 3 (Smooth discount curve)**.** The discount curve is sufficiently smooth in the sense that it has finite "roughness" under the smoothness measure

$$|g|_{\alpha,\delta} = \left(\int_0^\infty \left(\delta\, g'(x)^2 + (1-\delta)\, g''(x)^2\right) e^{\alpha x}\, dx \right)^{1/2},$$

where $\alpha \geq 0$ governs maturity weighting and $\delta \in [0,1]$ trades off penalizing slope versus curvature. Penalizing $g'(x)^2$ discourages oscillatory behavior ("tension"), while penalizing $g''(x)^2$ discourages sharp local kinks ("curvature"). This restriction should be interpreted as an economically motivated regularization device rather than as a literal description of the true data-generating process. Abrupt local changes in the discount curve can imply extreme payoffs for dynamic risk-free trading strategies; limits to arbitrage therefore favor sufficiently smooth curves.

Assumption 4 (Admissible function class)**.** The discount curve lies in the class $G_{\alpha,\delta}$ of twice weakly differentiable functions $g : [0,\infty) \to \mathbb{R}$ with $g(0) = 1$ and $|g|_{\alpha,\delta} < \infty$. The degenerate parameter combination $(\alpha, \delta) = (0, 0)$ is excluded to ensure existence and uniqueness of the solution. Additional mild boundary and regularity conditions apply in special cases (e.g., when $\delta = 0$ or $\delta = 1$) to guarantee that the norm and the implied kernel are well-defined.

3.4 Weighted fitting criterion and interpretation in yield units

Assumption 5 (Weighted least squares pricing fit)**.** The fit component of the estimator is based on weighted mean squared pricing errors,

$$\min_g \sum_{i=1}^{M} \omega_i \, (P_i - P_i^g)^2,$$

with exogenous weights $\omega_i > 0$.

Weights are chosen inversely proportional to squared price-scaled duration,

$$\omega_i = \frac{1}{M} \frac{1}{(D_i P_i)^2},$$

so that the weighted pricing error is approximately comparable to mean squared yield-fitting error,

$$\sum_{i=1}^{M} \omega_i (P_i - P_i^g)^2 \approx \frac{1}{M} \sum_{i=1}^{M} (Y_i - Y_i^g)^2,$$

where Y_i is the observed yield-to-maturity of bond i and Y_i^g is the yield-to-maturity implied by the model price P_i^g. This approximation motivates interpreting the criterion as operating in yield units rather than raw price units.

3.5 The regularized estimation problem

Putting these components together yields the core estimation problem: choose $g \in G_{\alpha,\delta}$ to trade off pricing fit against smoothness,

$$\min_{g \in G_{\alpha,\delta}} \left\{ \sum_{i=1}^{M} \omega_i \, (P_i - P_i^g)^2 + \lambda \, |g|_{\alpha,\delta}^2 \right\}, \qquad \lambda > 0.$$

The regularization parameter λ governs the balance between fitting observed bond prices and enforcing economically meaningful smoothness. In the reference approach, λ is selected empirically to control out-of-sample pricing performance.

4 Key Lemmas / Propositions Leading to Theorem 1

Theorem 1 (Filipović et al., 2022) delivers a closed-form kernel-ridge representation for the estimated discount curve. The result rests on a sequence of standard, but economically meaningful steps: (i) the pure pricing-fit problem is ill-posed without regularization, (ii) the smoothness penalty induces a Hilbert space structure with an associated reproducing kernel, and (iii) the representer theorem reduces the infinite-dimensional optimization problem to a finite-dimensional ridge regression. The reference paper develops these arguments rigorously using tools from functional analysis and reproducing kernel Hilbert spaces (RKHS). (Filipović et al., 2022) Here we summarize the key logical steps and provide intuition for each.

Lemma 1 (Ill-posedness without regularization)**.** *Consider the pure pricing-fit problem*

$$\min_g \sum_{i=1}^{M} \omega_i (P_i - P_i^g)^2$$

In typical Treasury applications, the number of observed securities M is substantially smaller than the number of maturity dates N required to represent a dense discount curve over long horizons (for example, $M \approx 300$ versus $N \approx 10{,}000$ for a daily grid over 30 years). As a consequence, the system of pricing equations is underdetermined, the curve g is not identified, and many distinct discount curves can fit observed prices similarly well.

Intuition. Observed bond prices impose only a limited set of linear restrictions on the discount curve. Without additional structure, the pricing equations leave a large null space, so regularization is not a numerical convenience but a structural requirement for identification.

Lemma 2 (Smoothness penalty induces a Hilbert space)**.** *Let $G_{\alpha,\delta}$ denote the class of twice weakly differentiable functions $g : [0, \infty) \to \mathbb{R}$ with $g(0) = 1$ and finite smoothness measure*

$$|g|_{\alpha,\delta} = \left(\int_0^\infty (\delta\, g'(x)^2 + (1-\delta)\, g''(x)^2) e^{\alpha x}\, dx \right)^{1/2}.$$

For $(\alpha, \delta) \neq (0, 0)$, this norm induces a Hilbert space structure on admissible discount curves (up to the normalization at $x = 0$).

Intuition. Penalizing derivatives constrains the "wiggliness" of the curve. The parameter δ shifts the penalty from curvature to slope ("tension"), while the exponential weight $e^{\alpha x}$ allows smoothness to vary with maturity. The Hilbert space structure is crucial for ensuring well-posedness and for applying kernel-based results.

Proposition 1 (Existence and uniqueness of the regularized minimizer)**.** *Consider the regularized objective*

$$\min_{g \in G_{\alpha,\delta}} \Big\{ \sum_{i=1}^{M} \omega_i (P_i - P_i^g)^2 + \lambda |g|_{\alpha,\delta}^2 \Big\}, \qquad \lambda > 0.$$

Once the smoothness penalty is included, the objective becomes strictly convex in g, and the minimization problem admits a unique solution $\hat{g}$.

Intuition. The term $\lambda|g|^2_{\alpha,\delta}$ eliminates flat directions that are left undetermined by the pricing equations alone, guaranteeing both existence and uniqueness of the estimator.

Proposition 2 (Representer theorem and finite-dimensional span)**.** *Because the smoothness norm induces an RKHS, the representer theorem applies: (Filipović et al., 2022) any minimizer of a regularized empirical objective over an RKHS can be written as a linear combination of the reproducing kernel evaluated at the relevant data locations. In the present setting, these locations are the cash-flow dates $x_1, \ldots, x_N$. Consequently, the solution takes the form*

$$\hat{g}(x) = 1 + \sum_{j=1}^{N} k(x, x_j)\beta_j,$$

for some coefficient vector $\beta \in \mathbb{R}^N$, where k is the kernel implied by (α, δ).

Intuition. Rather than selecting basis functions ad hoc, the smoothness penalty uniquely determines the functional basis through its associated kernel. Importantly, the effective dimension of the problem is driven by the number of cash-flow dates, not by an arbitrary maturity grid.

Proposition 3 (Kernel-ridge closed form and kernel specification)**.** *Let $K \in \mathbb{R}^{N\times N}$ be the kernel matrix with entries $K_{ij} = k(x_i, x_j)$, let $C \in \mathbb{R}^{M\times N}$ be the cash-flow matrix, and let $\Lambda = \mathrm{diag}(\lambda/\omega_1, \ldots, \lambda/\omega_M)$. Then Theorem 1 (Filipović et al., 2022) shows that the coefficient vector β is given by the kernel-ridge closed form*

$$\beta = C^\top (CKC^\top + \Lambda)^{-1}(P - C\mathbf{1}),$$

where the term $C\mathbf{1}$ reflects the normalization $g(0) = 1$. As a result, the estimated discount curve $\hat{g}$ is fully determined by linear algebra in dimension M once the kernel is specified.

Finally, Theorem 1 provides explicit expressions for the kernel $k(x, y)$ across the relevant parameter regimes, $\alpha = 0$ versus $\alpha > 0$ and $\delta \in (0, 1)$ versus $\delta \in \{0, 1\}$, and excludes the degenerate case $(\alpha, \delta) = (0, 0)$.

Intuition. The infinite-dimensional curve estimation problem collapses to a ridge regression in kernel space. The regularization parameter λ plays the role of ridge shrinkage, controlling the flexibility of the fitted discount curve.

5 Theorem 1 (Main Result in Our Scope)

The central theoretical result provides a closed-form solution to the regularized discount-curve estimation problem. (Filipović et al., 2022) Once the smoothness penalty is fixed, the optimal discount curve takes the form of a **kernel ridge regression** estimator: (Filipović et al., 2022) it lies in the linear span of kernel basis functions evaluated at the observed cash-flow dates, and its coefficients are obtained from a single linear-algebra operation. In particular, no numerical optimization over function spaces or ad hoc choice of basis functions is required.

Theorem 1 (Kernel-Ridge Solution)**.** *Consider the regularized estimation problem*

$$\min_{g \in G_{\alpha,\delta}} \left\{ \sum_{i=1}^{M} \omega_i \left(P_i - P_i^g\right)^2 + \lambda |g|_{\alpha,\delta}^2 \right\}, \qquad \lambda > 0,$$

where $P_i^g = \sum_{j=1}^{N} C_{ij} g(x_j)$*. Let* k *be the kernel induced by the norm* $|\cdot|_{\alpha,\delta}$*, and define the kernel matrix* $K \in \mathbb{R}^{N \times N}$ *by* $K_{ij} = k(x_i, x_j)$*. Let*

$$\Lambda = \operatorname{diag}(\lambda/\omega_1, \ldots, \lambda/\omega_M).$$

Then the problem has a ***unique*** *solution* $\hat{g}$*, given by*

$$\hat{g}(x) = 1 + \sum_{j=1}^{N} k(x, x_j)\beta_j, \tag{5.1}$$

where $\beta = (\beta_1, \ldots, \beta_N)^\top$ *is*

$$\beta = C^\top \left(CKC^\top + \Lambda\right)^{-1} (P - C\mathbf{1}). \tag{5.2}$$

The degenerate parameter pair $(\alpha, \delta) = (0, 0)$ *is excluded.*

5.1 Explicit kernel formulas

For any $x, y \geq 0$, write $x \wedge y = \min\{x, y\}$ and $x \vee y = \max\{x, y\}$. For completeness, the kernel $k(x, y)$ induced by the smoothness norm is given explicitly for the relevant parameter regimes.

(i) $\alpha = 0$**,** $\delta \in (0, 1)$**.**

$$k(x, y) = \frac{1}{\delta}(x \wedge y) + \frac{1}{2\delta\rho}\Big(e^{-\rho(x+y)} - e^{\rho(x\wedge y) - \rho(x \vee y)}\Big), \tag{5.3}$$

where

$$\rho = \sqrt{\frac{\delta}{1-\delta}}. \tag{5.4}$$

(ii) $\alpha = 0$**,** $\delta = 1$**.**

$$k(x, y) = x \wedge y. \tag{5.5}$$

(iii) $\alpha > 0$**,** $\delta = 0$**.**

$$k(x, y) = -\frac{x \wedge y}{\alpha^2} e^{-\alpha(x\wedge y)} + \frac{2}{\alpha^3}\Big(1 - e^{-\alpha(x\wedge y)}\Big) - \frac{x \wedge y}{\alpha^2} e^{-\alpha(x \vee y)}. \tag{5.6}$$

(iv) $\alpha > 0$**,** $\delta \in (0, 1)$**.**

$$\begin{aligned} k(x, y) = &- \frac{\alpha}{\delta\, \ell_2^2}\Big(1 - e^{-\ell_2 x} - e^{-\ell_2 y}\Big) + \frac{1}{\alpha\delta}\Big(1 - e^{-\alpha(x\wedge y)}\Big) \\ &+ \frac{1}{\delta\sqrt{D}} \left(\frac{\ell_1^2}{\ell_2^2} e^{-\ell_2(x+y)} - e^{-\ell_1(x\wedge y) - \ell_2(x \vee y)} \right), \end{aligned} \tag{5.7}$$

where

$$D = \alpha^2 + \frac{4\delta}{1-\delta}, \qquad \ell_1 = \frac{\alpha - \sqrt{D}}{2}, \qquad \ell_2 = \frac{\alpha + \sqrt{D}}{2}. \tag{5.8}$$

(v) $\alpha > 0$**,** $\delta = 1$**.**

$$k(x, y) = \frac{1}{\alpha}\Big(1 - e^{-\alpha(x \wedge y)}\Big). \tag{5.9}$$

5.2 Interpretation and implications

Although the object of interest is a function $g(\cdot)$ defined on a continuum of maturities, Theorem 1 shows that solving the regularized problem does not require searching over an infinite-dimensional space. The optimizer $\hat{g}$ lies in the finite-dimensional span generated by the kernel sections $\{k(\cdot, x_j)\}_{j=1}^N$ evaluated at the observed cash-flow dates. Economically, the data discipline the estimator through linear pricing restrictions at those dates, while the smoothness penalty determines how the curve is interpolated and extrapolated between and beyond them. (Filipović et al., 2022)

This separation of roles is central. Bond prices restrict the discount curve only at a limited set of maturities, leaving many degrees of freedom unresolved. The smoothness norm selects the economically most plausible completion of the curve among these observationally equivalent candidates. In this sense, the estimator combines exact pricing information where it exists with disciplined regularization where it does not.

From a computational perspective, the theorem implies substantial simplification. Once the cash-flow matrix C, the kernel matrix K, and the weighting and penalty matrix Λ are constructed, estimation reduces to solving a single linear system involving $CKC^\top + \Lambda$, which is of dimension $M \times M$. No numerical optimization over function spaces is required, making the estimator straightforward to implement and easy to reproduce.

The representation further implies that the estimated discount curve depends on the observed price vector P in an affine (linear plus constant) manner. This property is particularly important for simulation and inference: sampling variation in prices propagates through a fixed linear mapping determined by (C, K, Λ), rather than through a nested or iterative optimization routine. As a result, the estimator exhibits stable and transparent sensitivity to pricing noise.

Finally, the shape of the estimator is not chosen ad hoc. The economically motivated smoothness norm $|\cdot|_{\alpha,\delta}$ is the primitive modeling choice, and the kernel, and therefore the implied basis functions in Section 5.1 are mathematically determined by that choice. The regularization parameter λ plays a role analogous to ridge shrinkage: smaller values emphasize fit and flexibility, while larger values emphasize smoothness and stability. This fit–smoothness trade-off provides the organizing principle for the Monte Carlo evaluation and the empirical application that follow.

6 Implementation and Monte Carlo Design

This section describes the empirical implementation of the smoothness-regularized discount-curve estimator and the Monte Carlo design used to evaluate its finite-sample behavior. The analysis adopts a purely static, cross-sectional perspective. A single "true" discount curve is fixed on

a maturity grid, and repeated samples of noisy coupon-bond prices are generated from this curve using a fixed cash-flow structure. The objective is to assess how well the estimator recovers the underlying discount and yield curves in a controlled environment that isolates identification, regularization, and noise effects, without introducing time-series dynamics or forecasting considerations.

6.1 Calibration data and true discount curve

Let $y_0(\tau)$ denote the continuously-compounded zero-coupon yield at maturity τ. The corresponding true discount curve is

$$g_0(\tau) = \exp\big(-\tau\, y_0(\tau)\big).$$

The curve g_0 is held fixed across all Monte Carlo replications and provides the benchmark against which estimated curves are evaluated. The calibration step is not part of the estimation problem; it only fixes the term-structure shape used in the data-generating process.

6.2 Synthetic bond universe and cash-flow matrix

Let $0 < x_1 < \cdots < x_N$ denote a maturity grid consisting of semiannual points up to 30 years,

$$x_j \in \{0.5, 1.0, 1.5, \ldots, 30.0\}.$$

We construct a synthetic cross-section of coupon bonds with maturities $1, \ldots, 30$ years, coupon rates in

$$\{0, 1, 2, 3, 4, 5\}\% \quad \text{(annual rates)},$$

semiannual coupons, and face value 100. Coupon and principal payments are mapped to the grid and collected in a cash-flow matrix $C \in \mathbb{R}^{M\times N}$, where C_{ij} denotes the cash flow of bond i paid at date x_j. The maturity grid and cash-flow matrix are held fixed across replications.

Under the law of one price, the noise-free price vector implied by a candidate discount curve g is

$$P^g = Cg,$$

with $g = (g(x_1), \ldots, g(x_N))^\top$. In particular, the noise-free prices implied by the true curve are

$$P^0 = Cg_0.$$

6.3 Kernel bank and hyperparameter selection

Theorem 1 provides explicit kernel formulas for admissible parameter regimes (α, δ). In implementation, we do not fix a single regime ex ante. Instead, we construct a *kernel bank* containing: (i) the baseline family with $\alpha = 0$ and $\delta \in (0, 1)$, (ii) the limiting case $\alpha = 0, \delta = 1$, and (iii) the $\alpha > 0, \delta = 1$ family. Within each Monte Carlo replication, we select the kernel (regime and parameter value) and the ridge penalty λ by K-fold cross-validation using pricing RMSE as the objective.

6.4 Price generation, weighting, and estimation procedure

Observed bond prices are generated by perturbing noise-free prices with additive errors. For replication $b = 1, \ldots, B$,

$$P^{(b)} = P^0 + \varepsilon^{(b)}, \qquad \varepsilon^{(b)} \sim \mathcal{N}(0, \sigma^2 I_M),$$

independently across bonds and replications. The noise level is set as

$$\sigma = 0.05 \cdot \frac{1}{M} \sum_{i=1}^{M} P_i^0,$$

i.e., 5% of the mean noise-free bond price.

To emphasize pricing relevance and account for heterogeneous information content across bonds, we use duration-based weights computed under a baseline discount curve $g_{\text{base}}(\tau) = \exp(-\tau y_{\text{base}})$ with $y_{\text{base}} = 3\%$. Let D_i denote the Macaulay duration[1] of bond i computed under g_{base}. The weight for bond i in replication b is

$$\omega_i^{(b)} \propto \frac{1}{\left(D_i \, P_i^{(b)}\right)^2},$$

with a small floor to avoid numerical degeneracy. Both KR and NSS are tuned by the same K-fold cross-validation protocol using the weighted pricing RMSE objective. The reported *pricing RMSE* summaries below are computed as unweighted RMSE against the noise-free prices P^0 to remain interpretable in price units.

The KR estimator is implemented in a numerically stable baseline-centered form: we solve for deviations from g_{base} and set $\hat{g} = g_{\text{base}} + K\hat{\beta}$, where $\hat{\beta}$ is obtained from the closed-form kernel ridge system implied by Theorem 1.

6.5 Monte Carlo experiment and evaluation metrics

The Monte Carlo experiment consists of $B = 250$ independent replications. Across replications, g_0, $\{x_j\}_{j=1}^N$, and C remain fixed; only $\varepsilon^{(b)}$ varies.

We evaluate performance in price and yield space. Pricing accuracy is measured by unweighted RMSE relative to noise-free prices,

$$\text{RMSE}^{P,(b)} = \sqrt{\frac{1}{M} \sum_{i=1}^{M} \left((C\hat{g}^{(b)})_i - (Cg_0)_i\right)^2}.$$

Curve recovery is assessed via continuously-compounded zero-coupon yields

$$\hat{y}^{(b)}(x_j) = -\frac{1}{x_j} \log \hat{g}^{(b)}(x_j),$$

[1] Macaulay duration is the present-value–weighted average time to receipt of the bond's cash flows and measures its first-order sensitivity to parallel yield shifts.

and yield RMSE is reported in basis points and computed over maturities $x_j \geq 1$ year:

$$\mathrm{RMSE}_{\mathrm{bp}}^{y,(b)} = 10^4 \sqrt{\frac{1}{|\{j : x_j \geq 1\}|} \sum_{j:x_j \geq 1} \left(\hat{y}^{(b)}(x_j) - y_0(x_j)\right)^2}.$$

We additionally record numerical "bad-curve" flags (non-finite or implausibly large yields) to quantify estimator stability.

7 Monte Carlo Simulation I — Structural Validation (Our Model)

This section evaluates the smoothness-regularized FPY kernel-ridge (KR) estimator in the controlled cross-sectional environment of Section 6. The objective is structural validation: we assess whether the estimator recovers a stable and economically sensible discount and yield curve from coupon-bond prices subject to measurement noise.

7.1 Monte Carlo setup

We use the fixed true curve $g_0(\tau) = \exp\{-\tau y_0(\tau)\}$, a semiannual grid to 30 years, and a synthetic bond universe encoded in the cash-flow matrix C. In replication b, observed prices are

$$P^{(b)} = P_0 + \varepsilon^{(b)}, \qquad \varepsilon^{(b)} \sim \mathcal{N}(0, \sigma^2 I_M),$$

with σ calibrated as in Section 6. Hyperparameters (kernel choice from the bank and λ) are selected by K-fold cross-validation minimizing pricing RMSE, aligning tuning with the pricing role of the discount curve.

7.2 Evaluation metrics

We evaluate performance in price space and yield space. Pricing accuracy is measured by

$$\mathrm{RMSE}^{P,(b)} = \sqrt{\frac{1}{M} \sum_{i=1}^{M} \left((C\hat{g}^{(b)})_i - (Cg_0)_i\right)^2}.$$

Yield recovery is summarized in basis points over maturities $x_j \geq 1$ by

$$\mathrm{RMSE}_{\mathrm{bp}}^{y,(b)} = 10^4 \sqrt{\frac{1}{|\{j : x_j \geq 1\}|} \sum_{j:x_j \geq 1} \left(\hat{y}^{(b)}(x_j) - y_0(x_j)\right)^2}.$$

To summarize sampling uncertainty, we report distributional summaries across replications (Table A1) and visualize pointwise recovery bands for yields and discount factors (Figures A1–A3).

7.3 Baseline results: yield and discount curve recovery

Figure A1 reports Monte Carlo recovery bands for the yield curve. The median estimated yield curve tracks the true curve closely across maturities. Uncertainty is concentrated at very short

maturities, where identification from coupon-bond cash flows is intrinsically weaker and small perturbations in discount factors translate into larger yield movements.

Figure A2 makes this maturity dependence explicit by plotting yield errors $\hat{y}(x) - y_0(x)$ together with dispersion bands. Errors are centered near zero across maturities, while band widths vary systematically with identification strength implied by the cash-flow overlap structure.

Figure A3 provides the corresponding discount-factor recovery plot. Because coupon-bond prices are linear in discount factors, discount-factor recovery is directly tied to pricing accuracy. The figure shows smooth and stable discount-factor estimates that remain close to g_0.

7.4 Pricing accuracy and estimator stability

Pricing performance is assessed using global summaries and replication-level diagnostics. Table A1 reports the pricing RMSE distribution across $B = 250$ replications.

Figure A4 provides an intuitive diagnostic for one representative replication by plotting observed noisy prices against model-implied prices. Tight alignment around the 45° line indicates close reproduction of observed prices.

7.5 Summary

Overall, the Monte Carlo evidence provides structural validation for the KR estimator. In a coupon-bond environment with noisy prices, the estimator recovers a stable discount curve and delivers accurate pricing. Variation in yield recovery is largest at the short end, while medium and long maturities are estimated tightly.

8 Monte Carlo Simulation II — Benchmark Comparison Using Nelson–Siegel–Svensson

This section benchmarks the smoothness-regularized estimator against the Nelson–Siegel–Svensson (NSS) yield-curve specification under the same Monte Carlo design, cash-flow structure, and pricing-focused cross-validation protocol.

8.1 Benchmark specification and estimation

NSS represents the zero-coupon yield curve as

$$y(\tau) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} + \beta_2 \left(\frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} - e^{-\tau/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-\tau/\tau_2}}{\tau/\tau_2} - e^{-\tau/\tau_2} \right),$$

with implied discount curve $g(\tau) = \exp\{-\tau y(\tau)\}$. In each replication, NSS is tuned by K-fold cross-validation minimizing pricing RMSE. Within each fold, parameters are estimated by bounded quasi-Newton optimization with multiple random starts, and a ridge-type penalty on $(\beta_1, \beta_2, \beta_3)$ (with tuning parameter κ) is cross-validated jointly with the number of starts. (Gurkaynak et al., 2006)

8.2 Comparison with the smoothness-regularized estimator

The comparison is summarized in Table A1 and Figures A1–A4. Discount-factor recovery (Figure A3) is particularly informative because it reflects the pricing primitive. The replication-level pricing diagnostic (Figure A4) shows tighter alignment to the 45° line for KR than for NSS, consistent with the lower pricing RMSE distribution.

8.3 Interpretation and implications

Both estimators are tuned for pricing performance, but they embody different structural restrictions. NSS enforces a low-dimensional shape on yields, while KR estimates discount factors in an RKHS with smoothness regularization. (Filipović et al., 2022) In applications where internally consistent discounting and accurate pricing of coupon-bearing securities is the primary objective, pricing RMSE provides the natural evaluation lens, with yield recovery interpreted through maturity-specific diagnostics and the discount-factor perspective.

9 Results and Discussion

This section synthesizes the Monte Carlo evidence from Sections 7 and 8. All results are based on the same simulated price samples, the same cash-flow structure, and identical pricing-focused cross-validation for both methods.

9.1 Pricing accuracy and yield recovery

The Monte Carlo experiment consists of $B = 250$ independent replications. Pricing RMSE is the headline performance metric, as it coincides with the cross-validation objective and directly reflects the economic role of the discount curve as the pricing primitive for coupon bonds.

Across replications, the smoothness-regularized KR estimator achieves a mean pricing RMSE of 1.4588, compared to 1.9116 for NSS. The median pricing RMSE is 1.2969 for KR and 2.0120 for NSS. Thus, the reduction in pricing error is both economically and statistically meaningful, and it holds not only on average but also at the median. In distributional terms, the 10th and 90th percentiles of pricing RMSE are 0.9131 and 2.1538 for KR, versus 1.4002 and 2.3306 for NSS, indicating that KR dominates NSS across most of the distribution of simulation outcomes.

Yield recovery differences are more moderate in aggregate but still favor KR. The mean yield RMSE (measured in basis points and computed for maturities of at least one year) equals 31.8781 bp for KR and 33.8339 bp for NSS. The median yield RMSE is 23.8540 bp for KR and 28.3223 bp for NSS. Yield dispersion is higher under KR, with a standard deviation of 38.4801 bp compared to 13.9165 bp for NSS, reflecting greater variability in yield errors across replications. This dispersion underscores that scalar yield summaries conceal substantial maturity-specific heterogeneity.

Win-rate diagnostics reinforce the aggregate RMSE comparison. KR delivers a lower pricing RMSE than NSS in 89.6% of the 250 replications. Conditional on both estimators producing defined yield RMSE values, KR achieves a lower yield RMSE in 81.1% of comparable cases. Numerical stability diagnostics indicate that KR produces flagged “bad” curves in 4.8% of

replications, whereas NSS produces none under the adopted numerical guard. This pattern reflects the trade-off between flexibility and stability inherent in nonparametric regularization.

9.2 Economic interpretation

The maturity-specific error structure clarifies why yield-based summaries must be interpreted with care. Identification is weakest at very short maturities, where coupon-bond cash flows provide limited direct information and small perturbations in discount factors translate into amplified yield movements. At medium and long maturities, where cash-flow overlap is richer and information content is stronger, yield errors are more tightly centered around zero.

Because coupon-bond prices are linear in discount factors, discount-factor recovery is directly informative for pricing performance. The close alignment between estimated and true discount curves across most maturities is therefore consistent with the observed reduction in pricing RMSE under KR. In contrast, yield errors combine both discount-factor deviations and nonlinear transformation effects, which explains why yield RMSE improvements are smaller in magnitude than pricing RMSE improvements.

Overall, the pattern of results is consistent with the inverse problem interpretation of curve estimation. Pricing equations constrain the discount curve only at observed cash-flow dates, and smoothness regularization governs how the curve is interpolated between those dates. Where identification is strong, both KR and NSS perform well. Where identification is weak, the structure imposed by regularization becomes decisive.

9.3 Summary of findings

Three conclusions emerge from the Monte Carlo evidence. First, smoothness regularization materially improves pricing performance in a noisy coupon-bond environment, reducing mean pricing RMSE from 1.9116 under NSS to 1.4588 under KR. Second, yield accuracy is inherently maturity-dependent: while mean yield RMSE declines from 33.8339 bp to 31.8781 bp and median yield RMSE declines from 28.3223 bp to 23.8540 bp under KR, the dispersion of yield errors highlights the importance of examining maturity-specific diagnostics. Third, under identical pricing-focused tuning, the KR estimator dominates NSS in the vast majority of replications in both pricing and yield dimensions, at the cost of a small fraction of numerically unstable cases. Together, these findings support the view that economically motivated smoothness regularization provides a disciplined and effective framework for discount-curve estimation from coupon-bond prices.

10 Conclusion

This paper examined the smoothness-regularized framework of Filipović–Pelger–Ye for estimating the discount curve from coupon-bond prices in a static cross-sectional setting. (Filipović et al., 2022) The central theoretical insight is that recovering a continuous discount curve from a finite and noisy cross-section of coupon-bearing securities is an underdetermined inverse problem. Pricing equations alone do not uniquely identify the curve. Economically motivated smoothness regularization resolves this indeterminacy by restricting the admissible function

class, thereby converting an ill-posed problem into a uniquely solvable kernel-ridge system. Under the reproducing-kernel Hilbert space structure induced by the smoothness norm, the estimator admits a closed-form solution and can be implemented via a single linear system once the cash-flow matrix, kernel, and weighting structure are specified.

The Monte Carlo design isolates the core identification and regularization mechanisms in a controlled environment that mirrors key features of Treasury markets: overlapping coupon cash flows, heterogeneous maturities, and price-level noise. Across $B = 250$ independent replications, the kernel-ridge (KR) estimator recovers a stable and economically coherent discount curve. Because coupon-bond prices are linear in discount factors, discount-factor recovery is the economically primitive diagnostic. The simulations show close alignment between estimated and true discount curves across most maturities, with uncertainty concentrated at the short end where identification from coupon cash flows is intrinsically weak.

Benchmarking against Nelson–Siegel–Svensson (NSS) under identical simulated data and identical pricing-focused cross-validation highlights the quantitative implications of structural flexibility. The KR estimator reduces mean pricing RMSE from 1.912 under NSS to 1.459, a reduction of approximately 24%, and lowers the median pricing RMSE from 2.012 to 1.297. Yield accuracy improvements are more moderate but systematic: mean yield RMSE declines from 33.83 basis points to 31.88 basis points, and the median declines from 28.32 to 23.85 basis points. Win-rate diagnostics reinforce these results: KR delivers lower pricing RMSE in 89.6% of replications and lower yield RMSE in 81.1% of comparable cases. While KR exhibits higher dispersion in yield errors and produces flagged numerical instabilities in 4.8% of replications (reflecting greater functional flexibility in low-information regions), pricing performance remains consistently superior.

Three broader conclusions follow. First, smoothness regularization is not merely a technical device but an economically meaningful identification mechanism for discount-curve construction in coupon-bond markets. Second, evaluation metrics must align with economic purpose: when internally consistent discounting and accurate pricing are the primary objectives, pricing RMSE and discount-factor diagnostics provide the appropriate performance lens, while yield summaries should be interpreted jointly with maturity-specific error patterns. Third, under identical tuning protocols and data environments, a disciplined nonparametric framework can outperform a standard low-dimensional parametric benchmark in pricing accuracy, while highlighting the importance of numerical safeguards in flexible specifications.

Several extensions are natural. The simulation environment can be expanded to explore alternative noise intensities, maturity coverage patterns, and weighting schemes to map more precisely the regimes in which regularization delivers the largest gains. On the empirical side, applying the same cross-validation protocol to actual Treasury data would provide an external validity check and allow direct comparison of pricing errors across instruments. More generally, the results suggest that economically grounded regularization offers a principled and computationally tractable alternative to parametric yield specifications when the objective is coherent discount-curve estimation from coupon-bearing securities.

11 Additional Figures and Tables

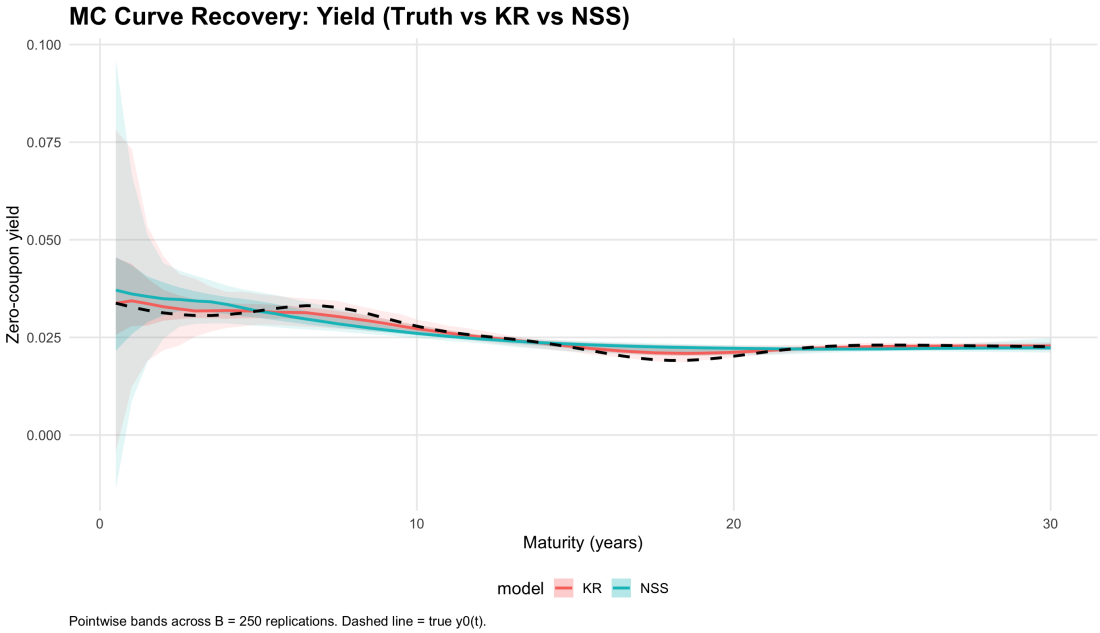

Figure A1: Monte Carlo yield curve recovery. Dashed line: true yield curve $y_0(t)$. Solid lines: median estimates across $B = 250$ replications. Shaded bands: pointwise dispersion across replications.

Table A1: RMSE summary across $B = 250$ Monte Carlo replications (pricing RMSE is the tuning and headline metric; yield RMSE is in basis points and computed for maturities ≥ 1 year).

Model	Metric	Mean	Median	SD	P10	P90
FPY (KR)	Pricing RMSE	1.458816	1.296861	0.493168	0.913054	2.153789
NSS	Pricing RMSE	1.911589	2.012030	0.359959	1.400155	2.330614
FPY (KR)	Yield RMSE (bp)	31.878147	23.853951	38.480123	14.002248	52.971056
NSS	Yield RMSE (bp)	33.833888	28.322324	13.916452	22.073203	53.718116

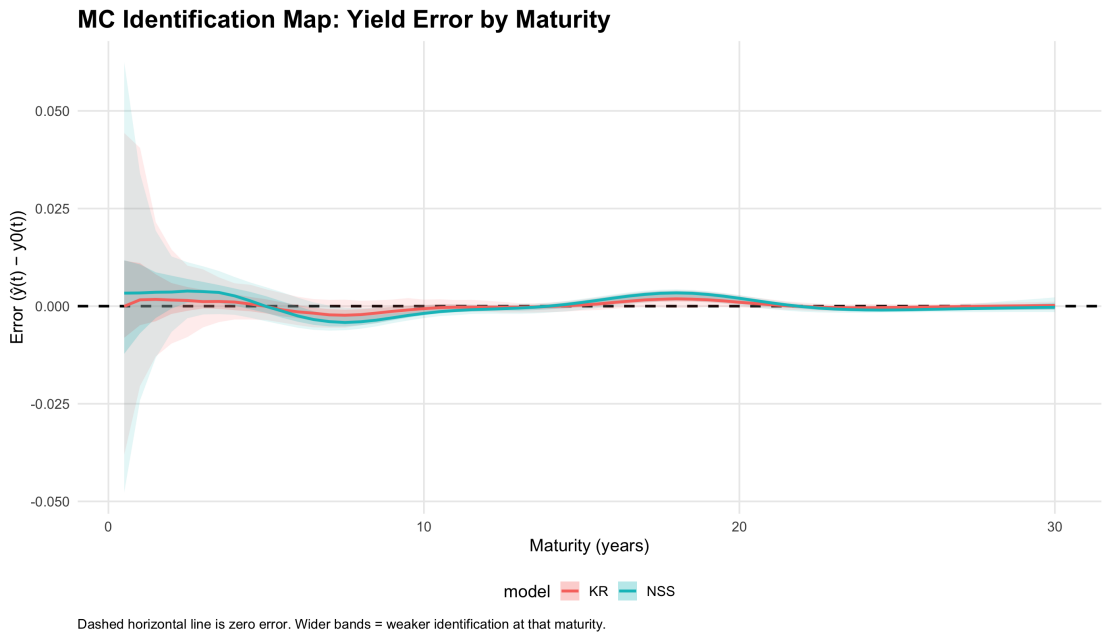

Figure A2: Identification map: yield error by maturity. Dashed horizontal line denotes zero error. Wider bands indicate weaker identification at that maturity.

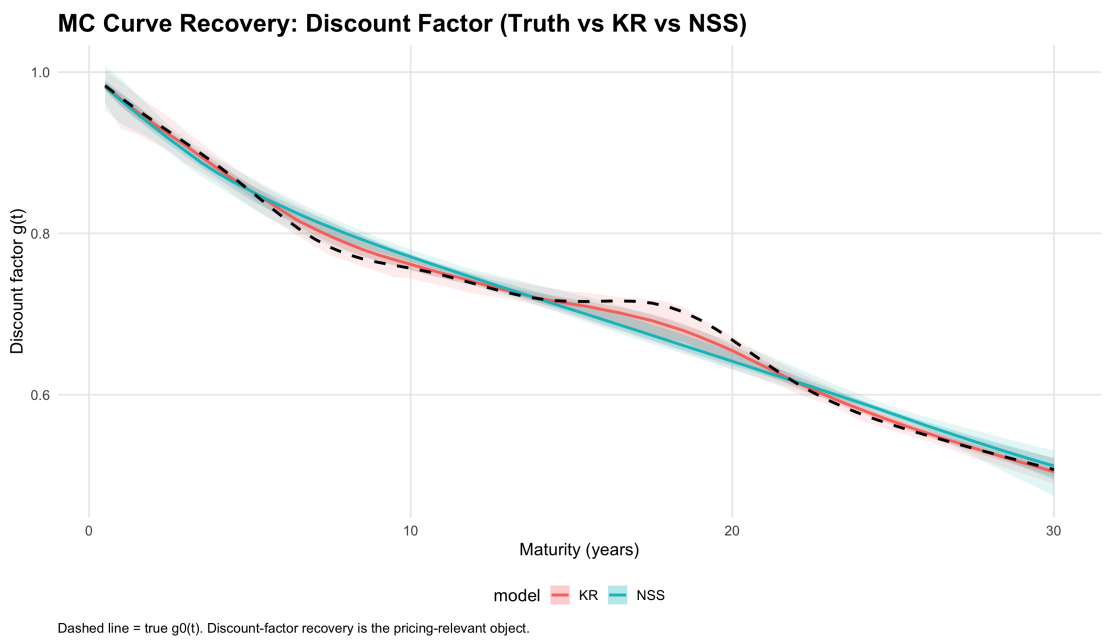

Figure A3: Monte Carlo discount-factor recovery. Dashed line: true discount curve $g_0(t)$. Solid lines: median estimates across $B = 250$ replications; shaded bands indicate pointwise dispersion.

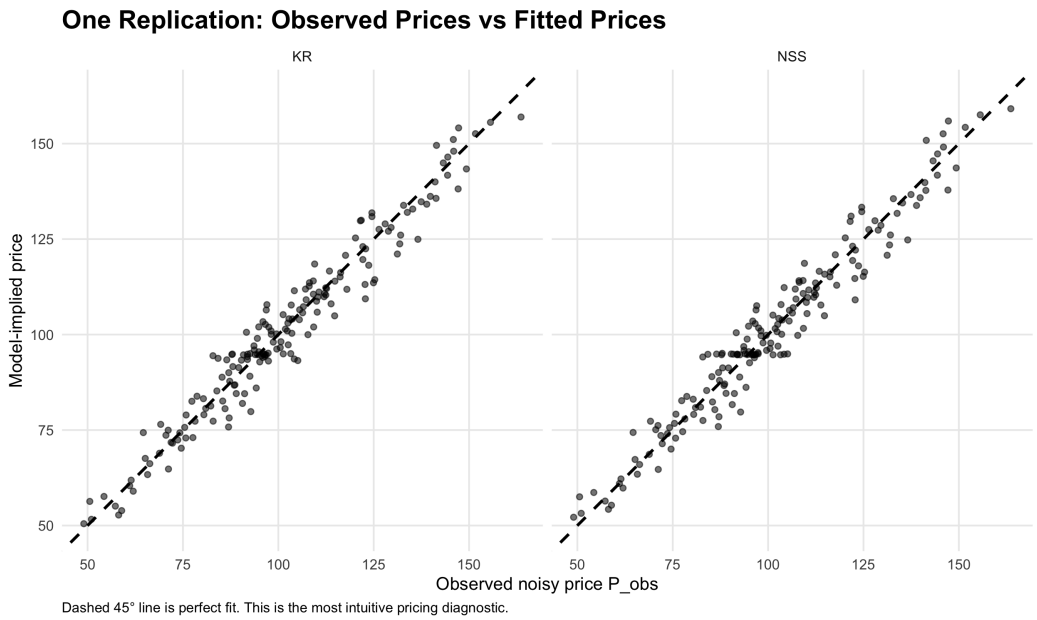

Figure A4: Observed versus fitted prices for one representative replication. Dashed 45° line indicates perfect fit. Left panel: KR (smoothness-regularized) fit. Right panel: NSS fit.

Model	Metric	Mean	Median	SD	P10	P90
FPY (KR)	Pricing RMSE	1.458816	1.296861	0.493168	0.913054	2.153789
NSS	Pricing RMSE	1.911589	2.012030	0.359959	1.400155	2.330614
FPY (KR)	Yield RMSE	31.878147	23.853951	38.480123	14.002248	52.971056
NSS	Yield RMSE	33.833888	28.322324	13.916452	22.073203	53.718116

Figure A5: Visual summary of RMSE distributional statistics across Monte Carlo replications (see Table A1 for numeric values).

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